Clim Dyn (2010) 35:143–157
DOI 10.1007/s00382-009-0597-5

# Scale decomposition of atmospheric water budget over West Africa during the monsoon 2006 from NCEP/GFS analyses

**Soline Bielli · Remy Roca**

Received: 19 December 2008 / Accepted: 17 May 2009 / Published online: 4 June 2009
© Springer-Verlag 2009

**Abstract** NCEP/GFS analysis is used to investigate the scale dependence and the interplay between the terms of the atmospheric water budget over West Africa using a dedicated decomposition methodology. The focus is on a 2-month period within the active monsoon period of 2006. Results show that the dominant scales of seasonal mean precipitation and moisture flux divergence over West Africa during the monsoon period are large scales (greater than 1,400 km) except over topography, where mean values of small scales (smaller than 900 km) are strong. Correlations between moisture flux divergences in monsoon and African Easterly Jet layers and precipitation indicate that precipitation is strongly correlated to moisture flux divergence via both large-scale and small-scale processes, but the correlation signal is quite different depending on the region and vertical layer considered. The analysis of the scales associated with the rainfall and the local evaporation over 3 different regions shows that positive correlation exists over the ocean between precipitation and evaporation especially at large scale. Over the continent south of the Sahel, the correlation is negative and driven by large scale. Over the northern part of Sahel, positive correlation is found, only at small scales during the active monsoon period. Lag correlation reveals that the maximum evaporation over the Sahel occurs 1–3 days after the maximum precipitation with maximum contribution from small-scale processes during the first day. This study shows that NCEP/GFS reproduces well the known atmospheric water budget features. It also reveals a new scale dependence of the relative role of each term of the atmospheric water budget. This indicates that such scale decomposition approach is helpful to clarify the functioning of the water cycle embedded in the monsoon system.

This paper is a contribution to the special issue on West African Climate, consisting of papers from the African Multidisciplinary Monsoon Analysis (AMMA) and West African Monsoon Modeling and Evaluation (WAMME) projects, and coordinated by Y. Xue and P. M. Ruti.

S. Bielli (✉)
Canadian Network for Regional Climate Modelling and Diagnostics, Université du Québec à Montréal @OURANOS, 550 Rue Sherbrooke Ouest, 19e, Montreal, QC H3A-1B9, Canada
e-mail: soline.bielli@cnrm.meteo.fr

S. Bielli · R. Roca
Laboratoire de Météorologie Dynamique, Tour 45-55, 3eme étage, Case Postale 99, 4 place Jussieu, 75252 Paris Cedex 05, France

## 1 Introduction

Rainfall broadly comes from the condensation of water vapor in the atmosphere. The water vapor distribution results from the balance between sources (surface evaporation and evapo-transpiration) and sinks (rainfall and clouds) as well as the transports between them. These elements are linked together through the atmospheric water cycle. When averaged over the globe over long time periods, rainfall equals evaporation. When a smaller region is considered, the recycling ratio (local evaporation to rainfall rate) departs significantly from a ratio of one and exhibits scale dependency (e.g., Eltahir and Bras 1996). This scale dependency is indicative of the complex interplays between both the non-local evaporation and moisture transport with local surface evaporation and rainfall. Understanding rainfall and its variability hence requires

characterizing the local versus the non local sources of moisture and their contributions to the water vapor distribution, in addition to characterizing the physical processes of vapor condensation (Trenberth et al. 2003). Over West Africa, the latter are upward motions which provide the needed pathway for condensation of vapor to rainfall. Water vapor distribution and rainfall are thus linked through large-scale upward motions and deep convection. The monsoon circulation induced rainfall over the Guinean Coast (large scale) is completed farther north by rainfall originating from organized deep convective systems (Dhonneur 1985).

The documented spatio-temporal variability of rainfall in West Africa is related to convection (e.g., Nicholson 1980; Janowiak 1988; Janicot 1992a, b, Rodwell and Hoskins 1996). In particular, it has been shown that 80% of rainfall over Sahel is associated with organized deep convective systems (e.g., Mathon and Laurent 2001). In this region, rainfall variability at the decadal scale is also related to the occurrence of individual systems (Le Barbe and Lebel 1997). These systems are part of a complex multiple scale interactions between convection and large-scale dynamics characterizing the monsoon (Redelsperger et al. 2002 and references therein). Strong scale dependences of the convective systems and size distributions have been highlighted at the scale of the African Easterly Waves (3–5 days) (Machado et al. 1993).

The link between rainfall and continental surface conditions takes place through local evaporation and water recycling. This link is rooted in surface memory effects through the land water cycle and the complex circulations amongst the ground reservoirs and vegetation. This yields to strong correlations between the conditions at one time and the rainfall at different space and time scales (Fontaine et al. 2003 and references therein). Over the West African basin, local evaporation accounts for one-third of rainfall at the seasonal scale (the other two-thirds being advected from the tropical Atlantic and Central Africa, e.g., Gong and Eltahir 1996). In the Sahel region, the Mediterranean area also appears as another source of non-local moisture (Fontaine et al. 2003; Nieto et al. 2006). The impact of both surface conditions and evaporation on rainfall has also been investigated at smaller scales. Eltahir and Pal (1996) and Taylor and Lebel (1998) showed that this link could be indirect and associated with mesoscale atmospheric circulations induced by surface condition anomalies. This calls for an analysis based on a scale decomposition of the rainfall/evaporation relationship over the West African continent, in order to sort out the relative contribution of the different processes involved. The complexity of the continental water budget further suggests a cautionary note on the background state of the continental reservoirs and their potential effects on the evaporation/rainfall relationship (Nicholson 2000). In order to consider an homogeneous response of the ground, this study is restricted to the active period of the rainy season.

The coupling between the large scale and synoptic circulations with the atmospheric water cycle in West Africa also calls for a scale decomposition approach to study the interactions between rainfall and water vapor transport. This need has already been identified as a mean to better understand rainfall variability (e.g., Desbois et al. 1988; Nicholson 2000), as well as to relate moisture advection, latent heat release and associated energy transports to the monsoon dynamics. Most previous studies over the region focused on the temporal scale of the monsoon dynamics and its signature on the water cycle (e.g. Kidson 1977; Lamb 1983; Cadet and Nnoli 1987; Long et al. 2000; Monkam 2007). In the present work, NCEP/GFS operational analyses are used to separate the spatial scales in evaporation, precipitation and moisture flux divergence fields in order to better identify the processes at play. The methodology used to decompose the regional-scale atmospheric water budget into different spatial scales has already been applied to Canadian Regional Climate Model (CRCM) output to study the atmospheric water budget over North America during both winter and summer seasons (Bielli and Laprise 2006, 2007). Here, the technique is adapted to the West African monsoon climate and, in particular, a layer selection is used to account for the basic dynamical elements of the region: the monsoon flux in the lower layer and the African Easterly Jet.

In the frame of the African Monsoon Multidisciplinary Analysis (AMMA) programme (Redelsperger et al. 2006), special observing periods took place during the 2006 monsoon season over West Africa. The analysis of these observations taken during this period is performed in details by many researchers having many different perspectives. The present work offers a large-scale framework for performing such an analysis (e.g. Janicot et al. 2008; Lebel et al. 2009). Summer 2006 shows a near-normal convective activity but with excess rainfall north of latitude 15N (Janicot et al. 2008). The dynamical onset of the monsoon occurred around June 25th followed by a transition period characterized by low convective activity, approximately until July 10th. Then, an active monsoon period characterized by a well-established monsoon and convective activity took place (Janicot et al. 2008). The latter period ranges approximately from 07/18/2006 to 09/14/2006, when the monsoon began to retreat towards the South (Thorncroft et al. 2007). Therefore, the focus here is on this two-month period associated with active monsoon conditions. The results from the analysis of this restricted time period should hence stand out as representative of active monsoon periods only. This period will be referred to as AM2006 thereafter.

The paper is organized as follows. Section 1 presents data and scale decomposition methodology. Section 2 contains the description of the mean atmospheric water budget for the AM2006 period. Section 3 completes the scale analysis with temporal correlations between scale decomposed moisture flux divergence, precipitation and evaporation to study the interplay between all the terms of the atmospheric water budget during the course of the monsoon. Conclusions, discussions and prospects are finally provided in Sect. 4.

## 2 Data and scale decomposition methodology

a) Atmospheric analyses and forecasts

The data used for this study come from the National Center for Environmental Prediction (NCEP) operational Global Forecast System (GFS) spectral model. The NCEP/GFS data were available every 6H on a 1° horizontal grid with 21 vertical pressure levels unevenly distributed from 1,000 hPa to 100 hPa. The data assimilation system used is the statistical spectral interpolation analysis scheme (SSI, Parrish and Derber 1992). Other details of the model are given in the NCEP Office Note 442 (2003). Daily averaged data are used over a subset domain [25W:25E, 10S:30N] covering West Africa during the AM2006 period. The data used from the NCEP/GFS analyses are: meridional and zonal wind components and specific humidity on pressure levels. Precipitation rate, latent heat flux and surface pressure are, on the other hand, taken from the 6-h forecast. Note that using the first guess humidity and dynamic fields instead of the analyzed fields does not improve nor deteriorate the closure of the water budget, so no extra bias is introduced in the results by mixing first guest and analyzed data. Evaporation is estimated by dividing the latent heat flux by the latent heat of vaporization ($L = 2.5e^6$ J/kg). The NCEP/GFS analysis have been chosen because of (i) their readily availability and (ii) quality compared to other datasets as experienced qualitatively by the authors on a near real time basis during the AMMA campaign monitoring (Lafore et al. 2006).

b) Rainfall estimates

The Global Precipitation Climatology Project (GPCP) one-degree daily (1dd) estimates of precipitation (Huffman et al. 2001) are used to evaluate the NCEP/GFS field. The GPCP data consists in the merging of satellite data (from both infrared and microwave imagers) together with rain gauges that are available from the Global Telecommunication Systems. Over West Africa during the monsoon, such product was shown to agree reasonably well with independent gauges from the Comité permanent Inter-états de Lutte contre la Sècheresse au Sahel for 10 days and longer time averages (Jobard et al. 2009). Hence, GPCP is here considered as a reference for the temporally averaged comparison with NCEP but no effort is attempted to use it in closing the water budget.

c) Scale decomposition methodology

Following Peixoto and Oort (1992), the atmospheric water budget can be written as:

$$\partial_t \overline{q} = -\nabla.\overline{\mathbf{Q}} + E - P \tag{1}$$

with the overbar representing vertical integration in pressure:

$$\overline{\psi} = \frac{1}{g}\int_{p_{\text{top}}}^{p_{\text{sfc}}} \psi \mathrm{d}p = \frac{1}{g}\int_{p_{\text{top}}}^{p_0} \beta\psi \mathrm{d}p \tag{2}$$

with $p_{\text{top}}$ and $p_{\text{sfc}}$ the lowest and the highest pressure values in a vertical atmospheric column, respectively. The term $\beta$ represents a mask to take into account the topography in the lower boundary (Boer 1982). $\mathbf{Q}$ is the horizontal moisture flux ($\mathbf{Q} = \mathbf{V}q$),$\mathbf{V}$ is the horizontal wind vector, $q$ is the specific humidity, $E$ is the evaporation rate and $P$ is the precipitation rate.

The scale decomposition based on Bielli and Laprise (2006, 2007) is adapted for the model, region and vertical structure of the atmosphere in West Africa during the monsoon season as follows. Each term $X$ of the water budget is decomposed into 2 spatial scales such that $X = X_L + X_S$ with the subscript $L$ representing large scales and the subscript $S$ small scales. The scale separation between large scales $L$ and small scales $S$ is performed by using the Discrete Cosine Transform (DCT, Denis et al. 2002). In Bielli and Laprise (2006, 2007), each element was decomposed into 3 scales, the large scales, the small scales and the third scale being the wave 0 (spatial average over the entire domain of interest). Here we use a slightly different decomposition (with only 2 bands, the band 0 representing the very large scale was not used as we are using a global model and thus assume that all the scales are resolved even if the decomposition is performed on a sub domain). The moisture flux divergence can thus be written as:

$$\bar{\mathbf{Q}} = \sum_{a,b} \overline{\mathbf{V}_a q_b} = \overline{\mathbf{V}_L q_L} + \overline{\mathbf{V}_L q_S} + \overline{\mathbf{V}_S q_L} + \overline{\mathbf{V}_S q_S} \tag{3}$$

with $(a, b) \in (L, S)$.

This decomposition allows taking advantage of the quadratic form of the moisture flux divergence and accessing the non-linear interactions between large and small scales. Following Bielli and Laprise (2007), these 4 terms are then recomposed into a large-scale term $(\nabla.\overline{\mathbf{Q}})_L$, which is related to the action of the large-scale wind on the

large-scale humidity, and a small-scale term $(\nabla.\overline{\mathbf{Q}})_S$, which is both associated with the small-scale terms and the non-linear interactions between large and small scales, such as:

$$\nabla.\overline{\mathbf{Q}} = (\nabla.\overline{\mathbf{Q}})_L + (\nabla.\overline{\mathbf{Q}})_S \quad (4)$$

with

$$(\nabla.\overline{\mathbf{Q}})_L = \nabla.\overline{\mathbf{V}_L q_L} \quad (4a)$$

$$(\nabla.\overline{\mathbf{Q}})_S = \nabla.\overline{\mathbf{V}_L q_S} + \nabla.\overline{\mathbf{V}_S q_L} + \nabla.\overline{\mathbf{V}_S q_S} \quad (4b)$$

Note that this decomposition is different from decomposing directly the moisture flux divergence into large and small scales. Indeed, as shown by Bielli and Laprise (2006) the interaction between large and small scales creates predominantly small scales. On the contrary, the self interaction of large scales creates both large and small scales that are comparable to the small scales created by the interaction of small and large scales. These results show that the relations $(\nabla.\overline{\mathbf{Q}})_L = E_L - P_L$ and $(\nabla.\overline{\mathbf{Q}})_S = E_S - P_S$ are not valid when using the moisture flux divergence decomposition 4a,b.

Moreover the moisture flux divergence is integrated over 3 layers: 1,000–800 hPa (monsoon layer or L1), 800–500 hPa (AEJ layer or L2) and 500–100 hPa (upper layer or L3) to closely follow the main vertical characteristics of the West African circulation. Vertical cross sections of mean and standard deviation of moisture flux divergence over the AM2006 period illustrates that the main moisture flux divergence activity is confined below 800 hPa in L1 where the low-level monsoon flow is confined (not shown). This layer decomposition is well adapted to study the water vapor fluxes over Africa (de Felice et al. 1982; Cadet and Nnoli 1987). The separation between L2 and L3 corresponds to the level above which the mean and standard deviation of moisture flux divergence is very weak. The height of the planetary boundary layer (HPBL) diagnosed in the NCEP/GFS analyses was not retained to select the layers for the vertical decomposition. Indeed, south of 10°N, HPBL is lower than the monsoon layer top, north of 20°N, HPBL is higher, and in between HPBL is located at about 800 hPa, corresponding to the level separating L1 and L2.

d) Selection of wavelength for the scale separation

Figure 1 displays the spatial variance spectra of precipitation for the stationary and the transient contributions calculated from the NCEP/GFS analysis over West Africa for the AM2006 period. The stationary spectrum is the spectrum of the mean field over the considered period. The transient spectrum is the mean of the spectra of the deviation fields (with respect to the temporal average over the AM2006 period) for each daily record. The precipitation

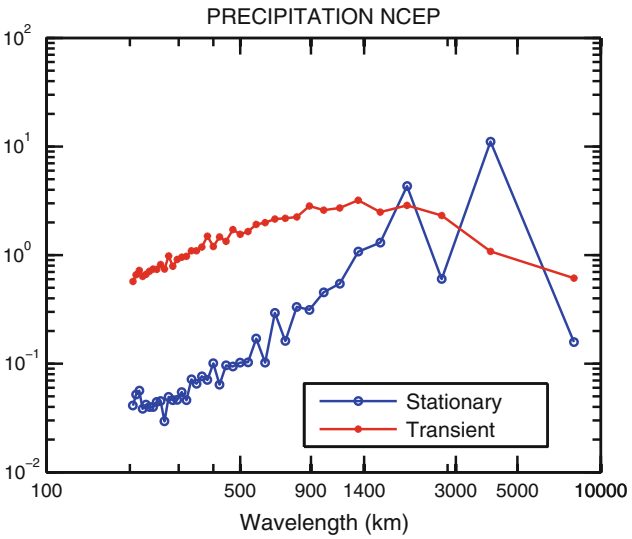



**Fig. 1** Stationary (*blue*) and transient (*red*) variance spectra of NCEP precipitation for the period 18 July 2006–14 September 2006 over West Africa

transient spectrum is close to a white spectrum and for wavelengths smaller than about 1,400 km, it is larger than the stationary part. For scales larger than 1,400 km, both transient and stationary spectra have about the same variance. The mean precipitation spectrum shows a break in its evolution that occurs near the 1,400 km wavelength.

In the following, all wavelengths greater than 1,350 km will be considered as large scales, and all wavelengths smaller than 900 km will be considered as small scales. In between, the filter follows a square cosine to minimize Gibbs effects. The upper range of the large scales that are resolved by the NCEP data within the domain of interest is around 8,000 km. This separation is further well suited for the African climate as it allows for separating the synoptic scales from the meso-alpha scales: following the decomposition, the classic synoptic African weather features associated with the African Easterly Waves (AEWs) corresponds to the large scale. The small scale on the other hand encompasses individual mesoscale convective systems and local circulation effects (sea breeze; mountain circulations; …).

e) Evaluation of NCEP/GFS precipitation using GPCP 1dd

Figure 2 shows the analysis of precipitation (mean and standard deviation) as inferred from NCEP/GFS analyses and GPCP 1dd observations for the AM2006 period. The red line corresponds to the mean position of the Inter Tropical Discontinuity (ITD) computed as the position of the 925-hPa dew point temperature being equal to 15°C. Computations are only performed for 00UTC when the ITD can be easily identified. This definition of the ITD

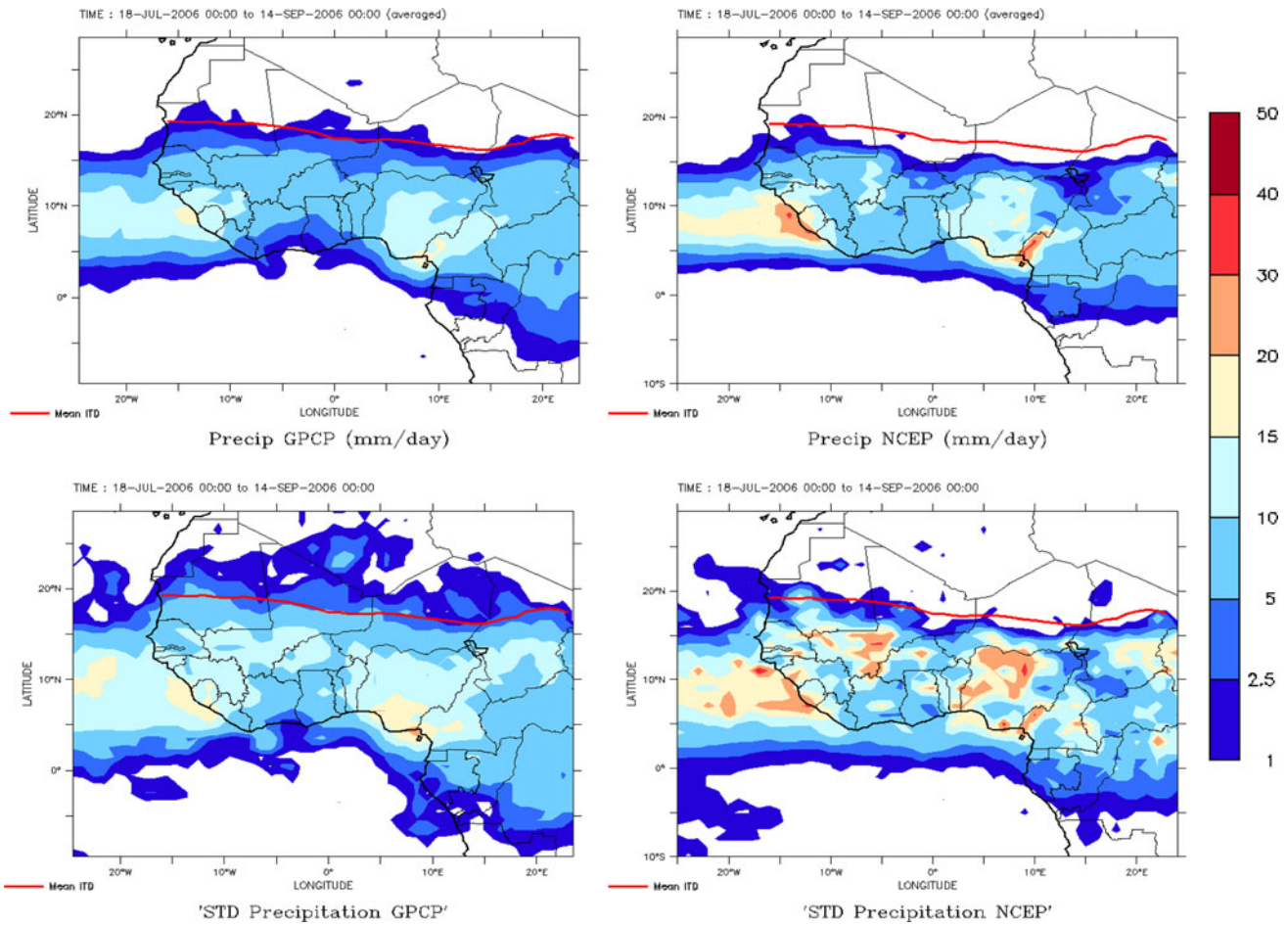



**Fig. 2** Mean (*top*) and standard deviation (*bottom*) of precipitation from daily values for the period 18 July 2006–14 September 2006 inferred from GPCP observations data (*left*) and from NCEP analyses (*right*). Values are in mm/day. The solid red line shows the mean

location was used during the intensive phase of the campaign by the forecasters of the African Center of Meteorological Application for Development (ACMAD) and is discussed at length in the Handbook of African Meteorology (Lafore et al. 2006). The ITD separates the two major regimes of the region: convection and rain occurs southwards, and environmental conditions are unfavourable to deep convection northwards.

The climatology of the summer precipitation over West Africa shows a zonal precipitation band with two regional precipitation maxima (Hastenrath 1994). One is centered on the West coast near 8°N and the second is located near the mountains of Cameroon, with a relative minimum in between. These climatological features are well captured by GPCP as previously discussed (Lamptey 2008). Overall, the NCEP/GFS analyses capture the two regions of maximum precipitation, and are able to reproduce the relative minimum in between, but with a general overestimation of precipitation compared to GPCP 1dd. Note that the northern extension of the rain band in the NCEP/GFS analysis appears limited with respect to the GPCP 1dd observations. Both standard deviation fields (NCEP/GFS and GPCP 1dd) show maximum variability associated with the precipitation maxima.

The total spectrum (stationary + transient) for GPCP 1dd precipitation is shown in Fig. 3 for comparison with NCEP/GFS. The two spectra are almost identical for wavelengths greater than about 1,400 km. For smaller wavelengths, NCEP/GFS exhibits more variance than GPCP 1dd. Note that 1,400 km is also the wavelength used to split between large scales and small scales. Both stationary and transient spectra for GPCP (not shown) have less variance than NCEP/GFS spectra for the small scales. This is consistent with the overall precipitation overestimation by NCEP/GFS both in term of mean and variability shown in Fig. 2. Hence, one must keep in mind for this present study that precipitation is generally overestimated (both in terms of mean and standard deviation) especially along the coast south of Dakar and along the coast of Cameroon in NCEP/GFS compared to GPCP 1dd.

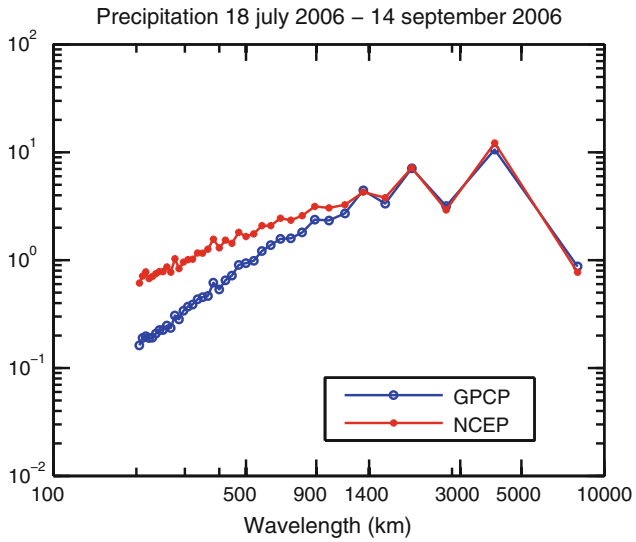



**Fig. 3** Total variance spectra for precipitation observed (GPCP, *blue line*) and from analyses (NCEP, *red line*) for the period 18 July 2006–14 September 2006

## 3 Mean water budget

The main balance of the atmospheric water budget is between precipitation and moisture flux divergence, with a small contribution from the evaporation mainly over the ocean. The mean moisture flux divergence displays a pattern very similar to that of precipitation with two convergence maxima separated by a local minimum. The mean precipitable water tendency term $[\partial_t \bar{q}]$ is negligible in the budget for a sufficiently long period like the one used here (i.e. 15 days and more, not shown).

Figure 4 presents the temporal mean values (top panels) of precipitation rate, evaporation rate, moisture flux divergence integrated over the monsoon layer and over the AEJ layer for the AM2006 period along with their large-scale (middle panels) and small-scale (bottom panels) contributions. Note that the third term of moisture flux divergence associated with the upper atmosphere (500–100 hPa) is not shown as water vapor is small in this layer and thus has a negligible contribution to the total budget. The solid red line corresponds to the mean position of the ITD and the dashed red lines show its minimum and maximum positions during the AM2006 period. The layer decomposition of the mean fields (Fig. 4, top panels) reveals two convergence bands in the monsoon layer (L1). The first band near the ITCZ around 10°N is associated with precipitation while near 20°N, it is mainly compensated by a divergence band in the AEJ layer (L2) corresponding to the low level dry circulation associated with the heat low (Peyrillé and Lafore 2007).

The dominant scale of the temporal mean atmospheric water budget is the large scale. Indeed, the evaporation term is large scale only. The mean moisture flux divergence is predominantly due to large scales, i.e. the action of the mean wind on the mean humidity. A number of features are nevertheless seen with significant contributions from the small scales, especially over the topography. This is true for both monsoon layer (L1) and AEJ layer (L2). Precipitation is also dominantly large scale except near topography (e.g. along the Guinean/Sierra Leone coast, near Mont Cameroon) where the small scale is almost as large as the large-scale part. Indeed, the precipitation field is characterized by a large band of precipitation with two main large-scale maxima of precipitation. The small scales are also confined over topographic regions, where they modulate the large-scale structures by reinforcing their central intensity and reducing their spatial extend, similarly to what was shown over North America in Bielli and Laprise (2006). The convergence band near 10°N in L1 also shows some small-scale features mostly due to topography but also along the coast, while the convergence/divergence bands near 20°N associated with the heat low are less influenced by the small scales and exhibit mainly a large scale signature. Chung et al. (2007) investigated the life cycle of deep convection and rainfall variability over West Africa using METEOSAT data and showed that organized, large convective perturbations that originally formed at smaller scale are associated with topography, confirming the precipitation scale dependence with topography found here.

In summary, large scale dominates the precipitation and the mean moisture flux divergence in both monsoon layer and AEJ layer except over topography (e.g. Sierra-Leone/Liberia and Nigeria/Cameroon) in monsoon layer, where the small-scale contribution is almost as large as the large-scale part. Similarly to what was shown by Bielli and Laprise (2006, 2007) for North America, the main stationary small-scale forcing of the moisture flux divergence in L1 is due to the topography. By construction, the small-scale contribution of the moisture flux divergence is the sum of three terms (cf. Eq. 4b). Amongst these three terms, the most important term is $\nabla.\overline{\mathbf{V}_S q_L}$ (not shown). This term is related to the action of the small-scale wind on the large-scale humidity field. This is different from North America where the dominant stationary small-scale term of the moisture flux divergence (both in winter and in summer) is the term involving large-scale wind and small-scale humidity (Bielli and Laprise 2006, 2007).

In complement to the mean water budget decomposition, the analysis of the variability of the budget terms (not shown) within the active monsoon period indicates that the variability of evaporation is negligible compare to the 3 other terms, the variability of the tendency of precipitable water is maximum in a zonal band around the ITD and it is mostly large scale. Finally, the maximum variability of

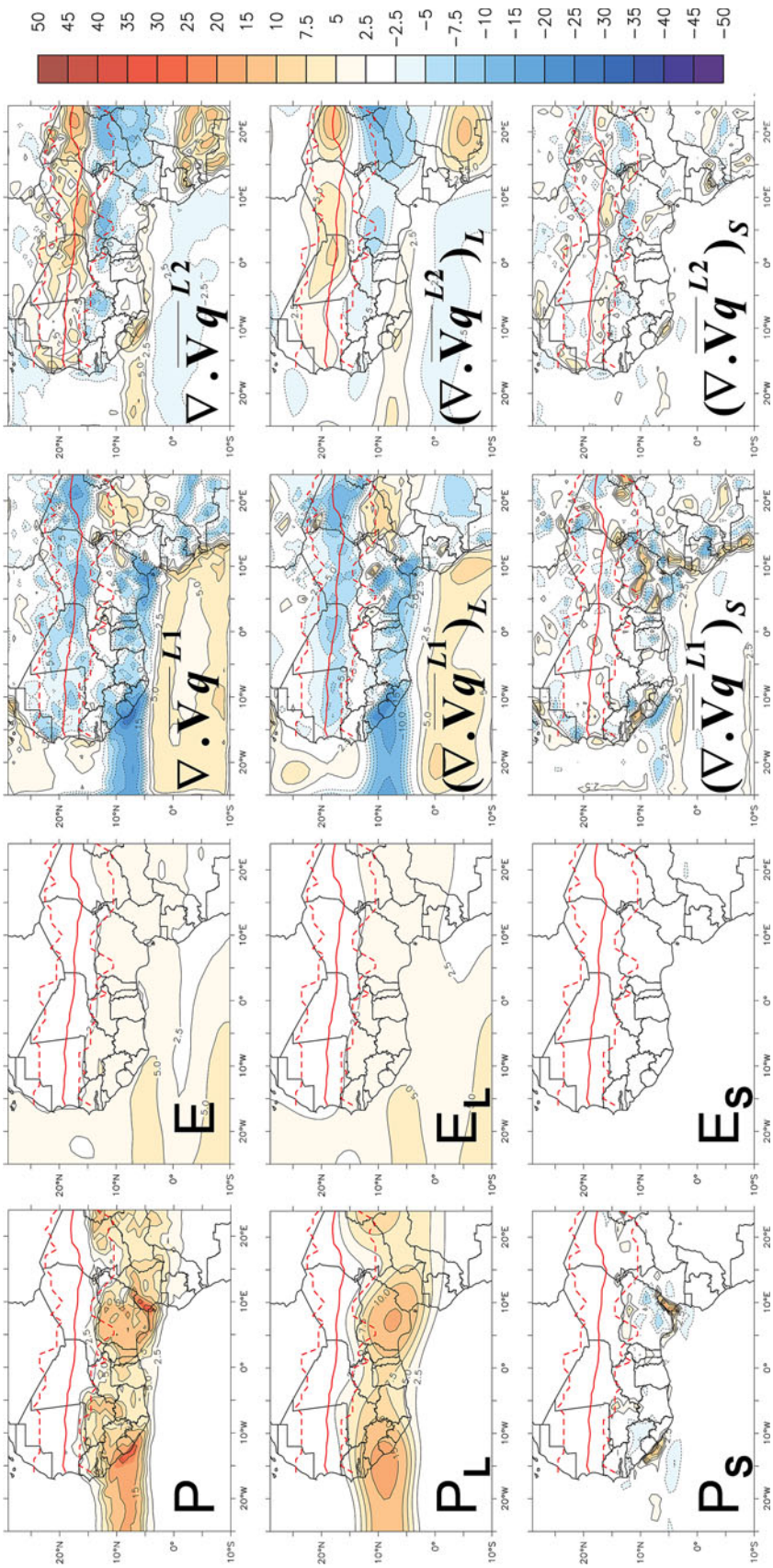



**Fig. 4** Mean atmospheric water budget (*top*) with its large-scale (subscript *L*, middle) and small-scale (subscript *S*, *bottom*) contribution over the period 18 July 2006–14 September 2006. *P* is for precipitation rate, *E* is the evaporation, $\nabla.\overline{\mathbf{V}q}^{L1}$ is the moisture flux divergence vertically integrated over monsoon layer and $\nabla.\overline{\mathbf{V}q}^{L2}$ is the moisture flux divergence vertically integrated over AEJ layer. $\partial_t \bar{q}$ the mean precipitable water tendency is not shown as it is negligible. Values are in mm/day, first contour is at ±2.5 mm/day. Negative (*blue*) values represent convergence zones and positive (*red*) values divergence zones. The solid red line shows the mean position of the ITD and the *dashed red lines* show the minimum and maximum positions

precipitation and moisture flux divergence in both layers (L1 and L2) occurs northward of the mean maxima, and small-scale variability is much larger than large-scale variability.

## 4 Interplay of the atmospheric water budget terms

The relationships among the individual atmospheric water budget terms are now further investigated by computing both spatial and temporal correlation coefficients.

a) Spatial analysis

Spatial correlation over the entire domain between temporal mean precipitation and temporal mean moisture flux divergence integrated over the entire atmosphere, over L1, L2 and L3 are listed in Table 1. Strong correlation links precipitation to vertically integrated moisture flux divergence with a value of almost –0.8. This relationship is strong for L1 ($R \sim -0.5$) but no correlation is found between precipitation and moisture flux divergence in L2. Correlation coefficients for the large-scale parts only are almost identical to those for the total fields, which is consistent with the fact that the temporal mean fields are dominated by the large-scale component. Spatial correlation coefficients between moisture flux divergence integrated over L1 and moisture flux divergence integrated over L2 for the total, large-scale and small-scale parts are respectively: −0.67, −0.58, −0.46. Both large-scale and small-scale processes play a role in the spatial correlation between L1 and L2 but with a slightly larger coefficient for the large scales. Spatial correlation between mean precipitation rate and mean evaporation rate for all scale is 0.4, for large-scale only 0.46 and for small scale only –0.11. This correlation is weaker than the correlation between precipitation and moisture flux divergence and it is correlated through large-scale terms only. This shows again the limited contribution of small-scale evaporation to the atmospheric water budget.

**Table 1** Spatial correlation coefficients between mean precipitation and mean moisture flux divergence during the active monsoon period 18/07/2006–14/09/2006

| | Column | L1 | L2 | L3 |
|---|---|---|---|---|
| $P$ | **−0.79** | **−0.51** | −0.12 | **+0.45** |
| $P_L$ | **−0.74** | **−0.51** | −0.14 | **+0.43** |
| $P_S$ | **−0.28** | −0.23 | −0.08 | **+0.37** |

The correlation is computed over the full domain shown in Fig. 3. L1, L2 and L3 refer respectively to the mean moisture flux divergence integrated over layer1, layer 2 and layer 3. The subscripts $L$ and $S$ refer respectively to large-scale and small-scale part of the fields. The correlations coefficients for $P_L$ and $P_S$ are calculated using only the large-scale part of precipitation and the large-scale part of moisture flux divergence

Bold values show correlations that are significant at the 95% confidence level

b) Precipitation and moisture flux divergence

Figure 5 displays maps of the temporal correlation coefficients between precipitation and moisture flux divergence over L1 and over L2. The correlation coefficients are computed only over the grid points for which more than 10 days having daily rainfall in excess of 1 mm/h are observed in order to focus on the region where the interplay of the water budget terms is not governed by single or too few rain events in the season. The analysis reveals that over the oceanic part of the rainfall band, the precipitation is mainly related to the convergence in the monsoon layer (Fig. 5a), with more homogenous areas for large-scale terms than for small-scale field, although both scales exhibit the same range of correlation coefficient. Above, at the AEJ level, oceanic precipitation correlates with the divergence in the layer although the related region is smaller than the previous one. Over land, precipitation is correlated with convergence in both monsoon and AEJ layer (Fig. 5b) with values greater than 0.75. Note that the $P$/L2 structure is slightly shifted to the north compared to the $P$/L1 correlation map. Note also that both large-scale and small-scale correlation coefficients between precipitation and moisture flux divergence in L1 and in L2 have about the same values. Finally the positive correlation over the ocean and along the coast between $P$ and L2 might be associated with shallow convection often observed over this region (Dhonneur 1985).

Precipitation also shows significant correlation with divergence in L3 (not shown) although it is associated with the small-scale processes. This is consistent with the deep convective nature of rainfall (small-scale processes in the present scale decomposition), having moisture convergence in the monsoon layer and divergence in the top layer.

Figure 5c displays temporal correlation coefficients between moisture flux divergences in monsoon and AEJ layers. It shows that south of the ITD between about 5°N and 15°N moisture flux divergence in the monsoon layer is not correlated with the moisture flux divergence in AEJ layer. Elsewhere, the two layers are strongly correlated through the small-scale fields. The strong small-scale correlation between the monsoon layer and the AEJ layer north of the ITD is not associated with convection but probably with the heat low structure (Chauvin et al. 2009). The large-scale correlation is significant only near the ITD and along the Gulf of Guinea.

In summary correlation between precipitation and moisture flux divergence in monsoon layer and AEJ layer is strong at all scales. Within the monsoon layer, the large-scale contribution dominates the correlation over the

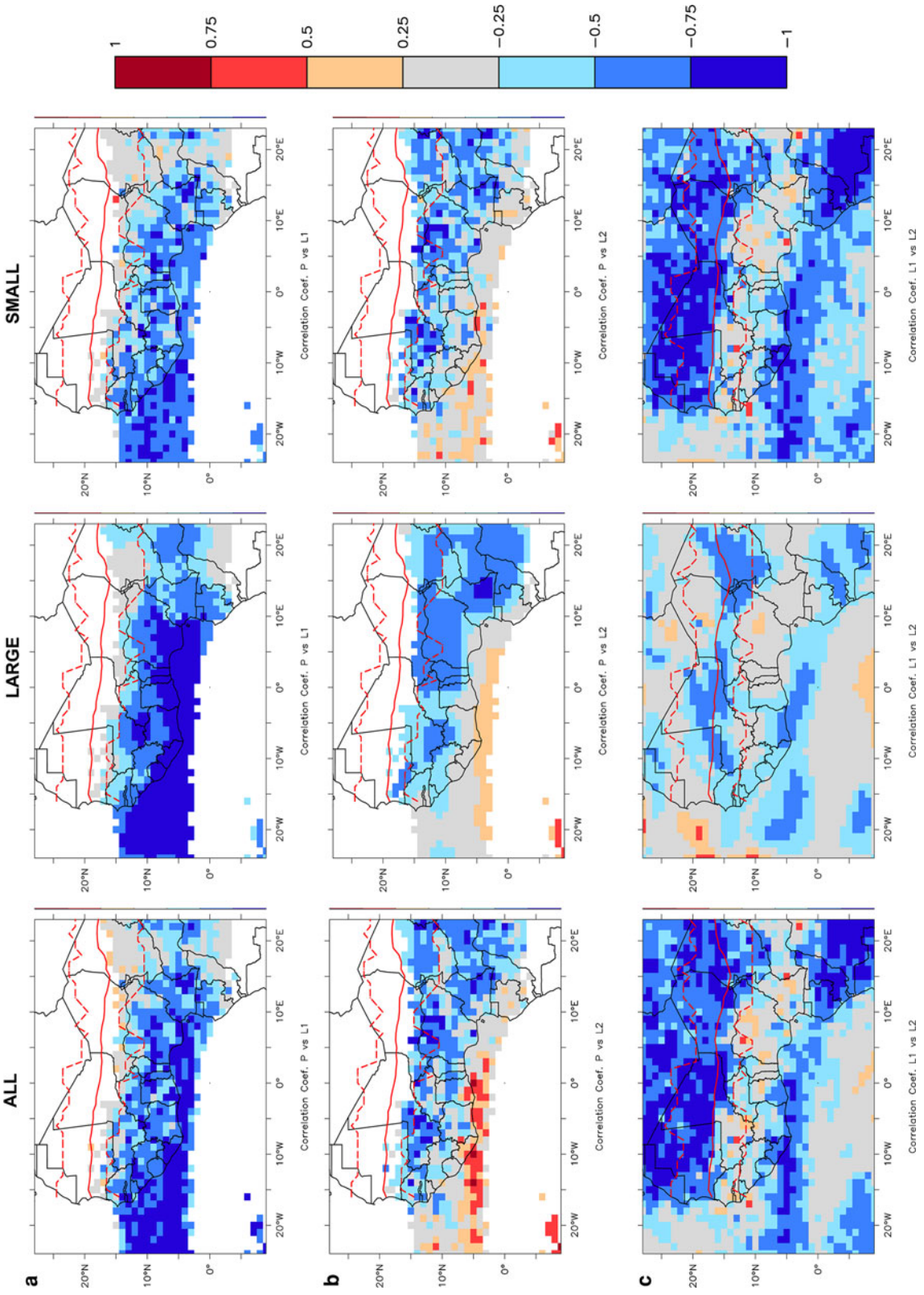



**Fig. 5** Temporal correlation coefficients between precipitation and moisture flux divergence integrated (**a**) over layer 1, (**b**) over layer 2, and (**c**) correlation between moisture flux divergence over layer 1 and moisture flux divergence over layer 2. *Left* for all scales, center: for large-scale part, and right: for small-scale part. Values of correlation greater than 0.25 and smaller than –0.25 are shown in grey and correspond to value below the 5% significance level. The *solid red line* shows the mean position of the ITD during the period, and the dashed lines show the minimum and maximum position of the ITD

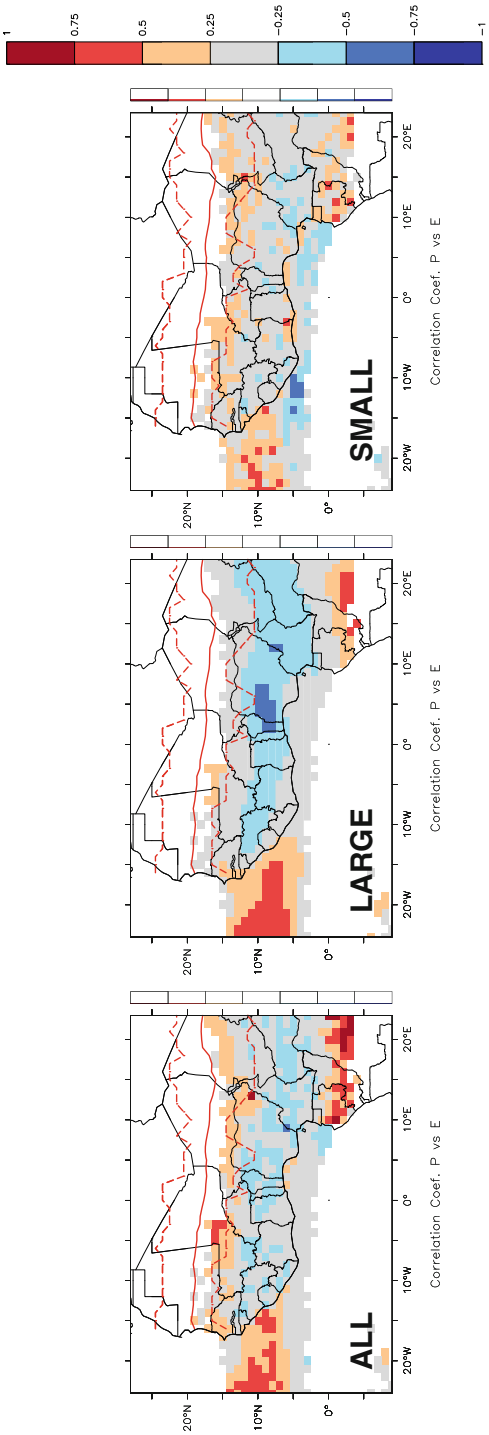



**Fig. 6** Temporal correlation coefficient between precipitation and evaporation for the period 18 July–14 September 2006 for the total fields (*left*), large-scale fields (*centre*) and small-scale fields (*right*). Only the points where more than 10 rain events greater than 1 mm/day are retained

ocean, while both large scales and small scales contribute to the correlation over the continent. In the AEJ layer, correlation is stronger and slightly more homogenous than in the monsoon layer over the Eastern part of the continent with equal values for large-scale and small-scale terms. Finally, there is no correlation between the monsoon layer and the AEJ layer in the Sahel region, and strong correlation northward and southward of the Sahel due to small-scale processes. The strong correlation to the North could be linked with the Heat Low structure, while the one to the South is probably related to the convergent monsoon flux and divergent AEJ.

c) Precipitation and evaporation

Figure 6 illustrates the positive correlation between precipitation rate and evaporation rate for all scales over the ocean near the coast of Senegal, in a zonal band directly south of the ITD, sporadically north of the southern bound of the ITD and south of the Equator over the continent in the Congo forests areas. This positive correlation is mainly due to the large-scale components over the ocean and over the Congo basin and to small-scale contributions over the northern part of Sahel.

In a continental zonal band between 0° and 10°N, the correlation between precipitation and evaporation is negative. In these areas where rain is frequent, the reservoirs are not the limiting factor to evaporation and increased precipitation does not imply increased evaporation. Indeed, this negative correlation might be in part due to a cooling associated with the precipitation and/or a moistening in the lower layers of the atmosphere, implying a decrease in the humidity gradient between the surface and the air just above that will limits the demand and decreases the evaporation rate. Figure 6 also indicates that this negative relationship between precipitation and evaporation south of 10°N is driven by the large-scale processes.

In summary, three distinct regions can be identified having 3 distinct behaviours: (1) the ocean region ([5N:12.5N, 25W:18W]) off Dakar coast where strong positive correlation occurs between precipitation and evaporation with contributions from both the large and the small scales; (2) the Guinean region ([5N:12.5N, 10W:24E]) along the coast of the Gulf of Guinea where negative correlation occurs between precipitation and evaporation. This correlation is mostly due to large-scale processes; (3) the Sahel region ([12.5N:17.5N, 15W:24E]) near the ITD where positive correlation occurs between precipitation and evaporation, largely due to small-scale processes the first day.

To further explore the precipitation and evaporation rates relationship over these typical regions, mean temporal lag correlation coefficients are calculated with a time lag varying between ±12 days. Positive/negative lag

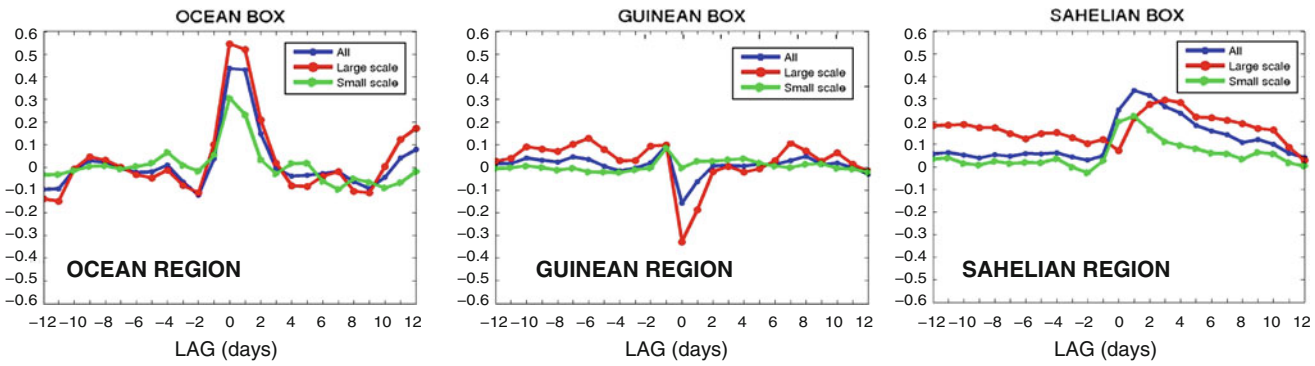



**Fig. 7** Temporal lag correlation (in day) between daily mean precipitation and daily mean evaporation. Positive/negative lag corresponds to precipitation leading/preceding evaporation. Mean values over 3 boxes: Ocean box: [5°N:12.5°N; 25°W:18°W]; Guinean box: [5°N:12.5°N; 10°W:24°E] and Sahelian box: [12.5°N:17.5°N; 15°W:24°E]. The *blue*, *red* and *green lines* display respectively the lag-correlation for all scales, large scales only and small scales only. 5% and 1% significance level correspond respectively to correlation of 0.25 and 0.33

corresponds to evaporation rate changing after/before precipitation rate. Figure 7 displays the results. The maximum correlation between $P$ and $E$ over the ocean occurs with a lag between 0 and 1 day over the ocean and no lag over the Guinean coast. These two curves present a well-defined and statistically significant peak of correlation, positive over the ocean box and negative over the Guinean box. In contrast, the maximum correlation over the Sahel box occurs with a positive lag of about one to 3 days (evaporation is larger 2 days after precipitation). Lag correlation of scale-decomposed fields confirms that correlation between $P$ and $E$ is largely dominated by large scale over the Guinean box with a coefficient greater than 0.4 for large-scale fields and no correlation for the small-scale field. Over the ocean, correlation between the large-scale fields is greater than correlation between small-scale fields, although this latter correlation is significant with a value of about 0.3. In both cases, large and small-scale correlations curves exhibit the same lag relationship. Over the Sahel box, the small-scale processes govern the correlation the first day and then large-scale processes become more important. At negative lag, the positive correlation between $P$ and $E$ over the Sahel box is not statistically significant at the 5% significance level.

## 5 Discussions and conclusions

The scale-decomposition methodology proposed by Bielli and Laprise (2006, 2007) to evaluate regional climate model simulations over North America is adapted to explore the scales involved in the atmospheric water budget over West Africa during the active monsoon period. The results concerning the analysis of the scales associated with the rainfall and the water vapor transport can be summarized as follows:

- Seasonal mean precipitation and moisture flux divergence structures in both monsoon and AEJ layers are dominated by the large scale components, except over high topography regions where small-scale and large-scale contributions are equivalent.
- Unlike previous analysis over the North American continent, the dominant term of the small scale moisture flux divergence is the term involving the transport of large-scale humidity by small-scale wind. This indicates that the features related to the small-scale divergence are mainly driven by mesoscale dynamics rather than by small-scale structures in the humidity field.
- Over the ocean, along the coastline, and over land along the Gulf of Guinea, precipitation is well correlated to moisture convergence in the monsoon layer, especially at large scale.
- Over the Sahel band, precipitation is correlated with the moisture convergence in the AEJ layer at all scales; no robust relationship can be identified between the moisture divergence in L1 and L2 over this area.

More generally, correlations between moisture flux divergences in L1, L2 and L3 reveal the existence of three distinct regions, which are all associated with a different behaviour: (1) South of the Equator we observe a positive correlation between the monsoon layer and the AEJ layer that is mainly associated with small scale processes. On the other hand, there is no correlation between the monsoon layer and the upper layer, as well as between the AEJ layer and the upper layer; (2) Between the Equator and the ITD, there is no correlation between the monsoon layer and the AEJ layer, and there is a negative correlation between both the AEJ layer and the monsoon layer and upper layer. The strongest correlation in this region is between AEJ layer and upper layer and for the small scales (not shown);

(3) North of the ITD there is a negative correlation between the monsoon layer and the AEJ layer, as well as between the AEJ layer and upper layer, and positive correlation between the monsoon layer and upper layer.

The results concerning the analysis of the scales associated with the rainfall and the local evaporative source can be summarized as follows:

- Precipitation and evaporation show positive correlation over the ocean, especially at large scales
- Over the wet continent, the relationship is negative and driven by large scale
- Over the northern part of Sahel, south of the ITD, positive correlation is observed, but only at small scales

Typical sub-regions have been selected to further analyse the interplay between the water budget terms. The oceanic region reveals a strong instantaneous (no-lag up to one day lag) correlation between rainfall and evaporation. The evaporation increase over the ocean associated with precipitation is due to an increase in surface winds that is seen at both small and large scales. Over the Guinean region, negative correlation is mainly associated with large-scale processes. This negative relationship might be in part due to rainfall induced surface cooling or surface air moistening which, therefore, decreases the gradient of humidity between the surface and the low level of the atmosphere, and by consequence, decreases evaporation. Therefore here, unlike over the semi-arid region, the wet surface conditions do not link an increase of precipitation with an increase of evaporation. Near and north of the ITD, the phase relationship is very different from the above-mentioned regions. The un-decomposed relationship suggests a smooth positive correlation between rainfall and evaporation with a temporal response of around one week, and a peak response around one day, which is in good agreement with the number usually quoted for Sahelian areas (e.g., Taylor and Lebel 1998). The scale decomposition further reveals that the signature of the small scales processes peaks over a shorter time window and precedes the large-scale processes signature that extends up to a week of influence. Our decomposition hence suggests that first, evaporation correlates with rainfall over Sahel at mesoscale (<900 km), followed by the larger scales (>1,400 km) which come later into play. The relationship between rainfall and local sources of moisture over the Sahel hence results from a mix between convective systems and synoptic weather patterns. Chauvin et al. (2005) used a General Circulation Model and performed sensitivity studies with and without constraints on the interaction between the dynamics and the soil moisture. Their composite analysis of the African Easterly Wave relationship between evaporation and dynamics showed that the control of the local evaporation is responsible for some of the atmospheric dynamical response. Their analysis revealed that even with such a short time frame as the AEW ones, the local source of evaporation was of importance to the wave dynamics and hence to the rainfall distribution. Our analysis may support such possible feedback between the local source of moisture, dynamics and rainfall over synoptic (between 1,400 km and 8,000 km) and time scales.

The results presented here are restricted to the active monsoon period. When the analysis is repeated before the onset, the interplay between the water budget elements remains similar over the ocean. Over the Guinean area, there is no more a relationship between evaporation and precipitation. Over the Sahel, there is no rainfall, so the correlation analysis cannot be applied. This, together with the above-summarized results, indicates that over the studied area different geographical rainfall regimes are related to different relationships with the transport of water vapor and the local sources, both in terms of temporal and spatial scales. In summary, the two sources of water vapor, local through evaporation and non-local through atmospheric transport, which are needed for rainfall production in West Africa, have been analysed. The scale dependence of the relative role of each source indicates that this kind of analysis should be of help to clarify the functioning of the water cycle embedded in the monsoon system. The scale dependence of the time at which the correlation between precipitation and evaporation peaks, as well as the regional sensitivity of this relationship, might be related to the different time scales of the components of the evaporation field (interception, soil evaporation and transpiration) as underscored by Scott et al. (1997). The elucidation of this should benefit from the numerous in situ hydrological and aerological measurements made during the AMMA campaign (Redelsperger et al. 2006).

The applications of this work are twofold: (i) strengthen and extend the validity of the present findings and (ii) use these results to question climate model representations of the interplay between the elements of the atmospheric water budget. One limitation of the present work arises from the use of a single season to establish the scale dependent lag relationship that might be influenced by the soil moisture state at the interannual timescale over West Africa, (Nicholson 2000; Mohr 2004) and seen in other regions as well (e.g., Bosilovich and Schubert 2001; Dirmeyer and Brubaker 2007). Alonge et al. (2007) indeed showed that different soil moisture regimes can modify the development of planetary boundary layer, and that wet regime creates a boundary layer that is more favourable to deep convection, hence modulating the time scale of the surface/precipitation relationship. Repeating our analysis over a longer time frame including contrasted monsoons at the interannual timescale could help to establish the

sensitivity of the scale dependence and of the lag relationship to the soil state at the beginning of the rainy season. Similarly, the atmospheric water budget has been derived from the NCEP/GFS analysis only, and our results may suffer from such a bias. The interplay between precipitation, moisture flux divergence over each layer and evaporation are probably strongly dependent on the convective scheme used in the model as well as the data assimilation procedure. Trenberth and Guillemot (1995) have shown that in the tropics, the initialization of the analyses as well as the physics of the assimilation model have an impact on the atmospheric water budget, especially through the large-scale divergence of moisture fields. Shay-El et al. (1999) have shown, using the NASA/GEOS analysis, that even the sign of the net budget of the semi-arid to arid regions, like the northern African region, is sensitive to the formulation of the assimilation procedure and to its intrinsic biases. Reapplying our method using a different analysis, like ECMWF or others, is one venue to tackle such potential sensitivity of our results. The AMMA reanalysis effort that will maximize the use of the enhanced observations acquired in 2006, including the dedicated surface budget term computations (de Rosnay et al. 2007) will also be of help toward such endeavour.

Finally, the representation of this kind of relationship in GCMs should be pursued. For instance, Gershunov and Roca (2004) showed that the simulated relationship between evaporation and the greenhouse effect over the Pacific Ocean using a coupled GCM was out of phase with the observed relationship. They suggested that such a bias was rooted in the parameterization of the model. Currently ongoing GCM inter-comparison exercises over West Africa (e.g., AMMA-MIP see Hourdin et al. 2008) will benefit from such an analysis that will confront the models with the analysis derived scale dependency lag relationships. Preliminary analyses are showing encouraging results with drastic differences in the representation of these relationships among the participating GCMs. The AMMA-MIP effort further provides a large database of model simulations with different surface and convective schemes. The analysis of these simulations along with the present study could highlight the dependence of our results to the formulation of the model's physics. The application of our technique to these lengthy simulations should further allow addressing the interannual variability sensitivity issues.

**Acknowledgments** This research was supported by the Canadian Foundation for Climate and Atmospheric Sciences (CFCAS) and the Fondation de l'Ecole Polytechnique. The first author is grateful to the Laboratoire de Météorologie Dynamique for the welcome as a visiting scientist. The authors thank Jean-Yves Grandpeix, Frédéric Hourdin and Jean-Philippe Lafore for very insightful and constructive discussions. Invaluable help with the GFS data from A. Deme is acknowledged. "Based on a French initiative, AMMA was built by an international scientific group and is currently funded by a large number of agencies, especially from France, UK, US and Africa. It has been the beneficiary of a major financial contribution from the European Community's Sixth Framework Research Program. Detailed information on scientific coordination and funding is available on the AMMA International web site http://www.ammainternational.org". The authors are also grateful to the two reviewers for their thoughtful and constructive reviews, which helped improving the manuscript.

## References

Lafore J-P et al (2006) Forecaster's guide for West African synthetic analysis/forecast WASA/F. AMMA EU Du2.1.a, 12 pp

Alonge CJ, Mohr KI, Tao WK (2007) Numerical studies of wet versus dry soil regimes in the West African Sahel. J Hydrometeorol 8:102–116. doi:10.1175/JHM559.1

Bielli S, Laprise R (2006) A methodology for regional scale-decomposed atmospheric water budget: application to a simulation of the Canadian Regional Climate Model nested by NCEP reanalyses over North America. Mon Weather Rev 134:854–873. doi:10.1175/MWR3098.1

Bielli S, Laprise R (2007) Time mean and variability of the scale-decomposed atmospheric water budget in a 25-year simulation of the Canadian Regional Climate Model over North America. Clim Dyn. doi:10.1007/s00382-007-0266-5

Boer GJ (1982) Diagnostic equations in isobaric coordinates. Mon Weather Rev 110:1801–1820. doi:10.1175/1520-0493(1982)110<1801:DEIIC>2.0.CO;2

Bosilovich MG, Schubert SD (2001) Precipitation recycling over the Central United States diagnosed from the GEOS-1 data assimilation system. J Hydrometeorol 2:26–35. doi:10.1175/1525-7541(2001)002<0026:PROTCU>2.0.CO;2

Cadet DL, Nnoli NO (1987) Water vapor transport over Africa and the Atlantic Ocean during summer 1979. Q J R Meteorol Soc 113:581–602. doi:10.1256/smsqj.47608

Chauvin F, Royer J-F, Douville H (2005) Interannual variability and predictability of African easterly waves in a GCM. Clim Dyn 24:523–544. doi:10.1007/s00382-004-0507-9

Chauvin F, Roehrig R, Lafore J-P (2009) Intraseaonal variability of the Saharan heat low and its link with the midlatitudes. J Clim (Submitted)

Chung ES, Sohn BJ, Schmetz J, Koenig M (2007) Diurnal variation of upper tropospheric humidity and its relation to convective activities over tropical Africa. Atmos Chem Phys 7:2489–2502

De Felice P, Viltard A, Camara M (1982) Vapeur d'eau dans la troposphère en Afrique de l'Ouest. Meteorologie 29:129–134

de Rosnay P, Boone A, Beljaars A, Polcher J (2007) AMMA land-surface modeling and intercomparison projects. GEWEX News 16(1):10–11

Denis B, Côté J, Laprise R (2002) Spectral decomposition of two-dimensional atmospheric fields on limited-area domains using the discrete cosine transform (DCT). Mon Weather Rev 130:1812–1829. doi:10.1175/1520-0493(2002)130<1812:SDOTDA>2.0.CO;2

Desbois M, Kayiranga T, Gnamien B, Guessous S, Picon L (1988) Characterization of some elements of the Sahelian climate and their interannual variations for July 1983, 1984 and 1985 from the analysis of Meteosat ISCCP data. J Clim 1:867–904. doi:10.1175/1520-0442(1988)001<0867:COSEOT>2.0.CO;2

Dhonneur G (1985) Traité de météorologie tropicale: application au cas particulier de l'Afrique occidentale et centrale, 150 pp, Eds : Météo-France, Paris

Dirmeyer PA, Brubaker KL (2007) Characterization of the global hydrological cycle from back-trajectory analysis of atmospheric water vapor. J Hydrometeorol 8:20–37. doi:10.1175/JHM557.1

Eltahir EAB, Bras RL (1996) Precipitation recycling. Rev Geophys 34(3):367–379. doi:10.1029/96RG01927

Eltahir EAB, Pal JS (1996) Relationship between surface conditions and subsequent rainfall in convective storms. J Geophys Res 101:26237–26245. doi:10.1029/96JD01380

Fontaine B, Roucou P, Trzaska S (2003) Atmospheric water cycle and moisture fluxes in the West African monsoon: mean annual cycles and relationship using NCEP/NCAR reanalysis. Geophys Res Lett 30(3). doi:10.1029/2002GL015834

Gershunov A, Roca R (2004) Coupling of latent heat flux and the greenhouse effect by large-scale tropical/subtropical dynamics diagnosed in a set of observations and model simulations. Clim Dyn 22:205–222. doi:10.1007/s00382-003-0376-7

Gong C, Eltahir E (1996) Sources of moisture for rainfall in West Africa. Water Resour Res 32:3115–3121. doi:10.1029/96WR01940

Hastenrath S (1994) Climate dynamics of the tropics. An updated edition of climate and circulations of the tropics. Kluwer Academic Publishers, Norwell, p 488

Hourdin F, Musat I, Guichard F, Ruti PM, Favot F, Filiberti M-A, Pham M, Grandpeix J-Y et al (2008) AMMA-Model Intercomparison Project. Bull Am Meteorol Soc (submitted)

Huffman GJ, Adler RF, Morrissey MM, Bolvin DT, Curtis S, Joyce R, McGavock B, Susking J (2001) Global precipitation at one-degree daily resolution from multisatellite observations. J Hydrometeorol 2:36–50. doi:10.1175/1525-7541(2001)002<0036:GPAODD>2.0.CO;2

Janicot S (1992a) Spatiotemporal variability of West African rainfall. Part I: regionalizations and typings. J Clim 5:489–497. doi:10.1175/1520-0442(1992)005<0489:SVOWAR>2.0.CO;2

Janicot S (1992b) Spatiotemporal variability of West African rainfall. Part II: associated surface and airmass characteristics. J Clim 5:499–511. doi:10.1175/1520-0442(1992)005<0499:SVOWAR>2.0.CO;2

Janicot S, Thorncroft C, Ali A, Asencio N, Berry G, Bock O, Bourles B, Caniaux G, Chauvin F, Deme A, Kergoat L, Lafore J-P, Lavaysse C, Lebel T, Marticorena B, Mounier F, Nedelec P, Redelsperger J-L, Ravegnani F, Reeves CE, Roca R, de Rosnay P, Schlager H, Sultan B, Tomasini M, Ulanovsky A (2008) Large-scale overview of the summer monsoon over West Africa during the AMMA field experiment in 2006. Ann Geophys 26:2569–2595

Janowiak JE (1988) An investigation of interannual rainfall variability in Africa. J Clim 1:240–255. doi:10.1175/1520-0442(1988)001<0240:AIOIRV>2.0.CO;2

Jobard I, Chopin F, Berges JC, Roca R (2009) An intercomparison of 10-day precipitation satellite products during West African monsoon. QJRMS (Submitted)

Kidson JW (1977) African rainfall and its relation to upper air circulation. Q J R Meteorol Soc 103:441–456. doi:10.1002/qj.49710343705

Lamb PJ (1983) West African water vapor variations between recent contrasting sub-Saharan rainy seasons. Tellus 30:240–251

Lamptey BL (2008) Comparison of gridded multisatellite rainfall estimates with gridded gauge rainfall over West Africa. J Appl Meteorol Climatol 47:185–205. doi:10.1175/2007JAMC1586.1

Le Barbe L, Lebel T (1997) Rainfall climatology of the HAPEX-Sahel region during the years 1950–1990. J Hydrol (Amst) 189:43–73. doi:10.1016/S0022-1694(96)03154-X

Lebel T, Parker TJ, Flamant C, Bourles B, Marticorena B, Mougin E, Peugeot C, Diedhiou A, Ngamini JB, Polcher J, Redelsperger JL, Thorncrof CD (2009) The AMMA field campaigns: multiscale and multidisciplinary observations in the West African region, submitted to QJRMS Special Issue on AMMA SOP, 2009

Long M, Entekhabi D, Nicholson SE (2000) Interannual variability in rainfall, water vapor flux, and vertical motion over West Africa. J Clim 13:3827–3841. doi:10.1175/1520-0442(2000)013<3827:IVIRWV>2.0.CO;2

Machado LT, Duvel JP, Desbois M (1993) Diurnal variations and modulation by easterly waves of the size distribution of convective cloud clusters over West Africa and the Atlantic Ocean. Mon Weather Rev 121:37–49. doi:10.1175/1520-0493(1993)121<0037:DVAMBE>2.0.CO;2

Mathon V, Laurent H (2001) Life cycle of Sahelian mesoscale convective cloud systems. Q J R Meteorol Soc 127:377–406. doi:10.1002/qj.49712757208

Mohr KI (2004) Interannual, monthly, and regional variability in the wet season diurnal cycle of precipitation in sub-Saharan Africa. J Clim 17:2441–2453. doi:10.1175/1520-0442(2004)017<2441:IMARVI>2.0.CO;2

Monkam D (2007) Influence des deux régimes d'ondes d'est sur les flux de vapeur d'eau en Afrique de l'Ouest durant l'été 1981. Secheresse 18(2):89–95

NCEP Office Note 442 (2003) The GFS Atmospheric Model, 14. Can be obtained from www.weather.gov/ost/climate/STIP/AGFS_DOC_1103.pdf

Nicholson SE (1980) The nature of rainfall fluctuations in subtropical West Africa. Mon Weather Rev 108:473–487. doi:10.1175/1520-0493(1980)108<0473:TNORFI>2.0.CO;2

Nicholson S (2000) Land surface processes and sahel climate. Rev Geophys 38(1):117–139. doi:10.1029/1999RG900014

Nieto R, Gimeno L, Trigo RM (2006) A Lagrangian identification of major sources of Sahel moisture. Geophys Res Lett 33:L18707. doi:10.1029/2006GL027232

Parrish DF, Derber JC (1992) The National Meteorological Center's spectral statistical interpolation analysis system. Mon Weather Rev 120:1747–1763. doi:10.1175/1520-0493(1992)120<1747:TNMCSS>2.0.CO;2

Peixoto JP, Oort AH (1992) Physics of climate. American Institute of Physics, New York, p 520

Peyrillé P, Lafore J-P (2007) An idealized two-dimensional framework to study the West African monsoon. Part II: large-scale advection and the diurnal cycle. J Atmos Sci 64:2783–2803. doi:10.1175/JAS4052.1

Redelsperger J-L, Diongue A, Diedhiou A, Ceron J-P, Diop M, Gueremy J-F, Lafore J-P (2002) Multi-scale description of a Sahelian synoptic weather system representative of the West African monsoon. Q J R Meteorol Soc 128:1229–1257. doi:10.1256/003590002320373274

Redelsperger J-L, Thorncroft CD, Diedhiou A, Lebel T, Parker DJ, Polcher J (2006) African monsoon multidisciplinary analysis—an international research project and field campaign. Bull Am Meteorol Soc 87:1739–1746. doi:10.1175/BAMS-87-12-1739

Rodwell MJ, Hoskins BJ (1996) Monsoons and the dynamics of deserts. Q J R Meteorol Soc 122:1385–1404. doi:10.1002/qj.49712253408

Scott R, Entekhabi D, Koster R, Suarez M (1997) Timescales of land surface evapotranspiration response. J Clim 10:559–566. doi:10.1175/1520-0442(1997)010<0559:TOLSER>2.0.CO;2

Shay-El Y, Alpert P, Da Silva A (1999) Reassessment of the moisture source over the Sahara desert based on NASA reanalysis. J Geophys Res 104:2015–2030. doi:10.1029/1998JD200003

Taylor CM, Lebel T (1998) Observational evidence of persistent convective-scale rainfall patterns. Mon Weather Rev 126:1597–1607. doi:10.1175/1520-0493(1998)126<1597:OEOPCS>2.0.CO;2

Thorncroft CD, Lafore J-P, Berry G, Roca R, Guichard F, Tomasini M, Asencio N (2007) Overview of African systems during the summer 2006, vol 12. No. 2, April 2007, CLIVAR Newsletter Exchanges

Trenberth KE, Guillemot CJ (1995) Evaluation of the global atmospheric moisture budget as seen from analyses. J Clim 8:2255–2272. doi: 10.1175/1520-0442(1995)008<2255:EOTGAM>2.0.CO;2

Trenberth KE, Dai A, Rasmussen RM, Parsons DB (2003) The changing character of precipitation. Bull Am Meteorol Soc 84:1205–1217. doi:10.1175/BAMS-84-9-1205